香港城市大學
City University of Hong Kong
Innovating into the Future

Tutorial 5: Simulator

Simulators for AD Systems

- Traditionally, perception data generated by manually labelling
  - E.g., the 3D position and boundaries of obstacles, vehicles, and lane markings …
- This is costly, time-consuming, and error-prone.
- Simulators enable to do this labor-intensive task in a virtual environment
  - Simulators provide **quicker, cheaper** and **safer** tools compared with real environment

Scene in simulator and real footage

Simulators for AD Systems

- Apollo Simulator is built on Unity
- CARLA is built on Unreal Engine 5
  - CARLA, a mainstream opensource simulator for AD systems

Apollo Simulator

Data collected from Apollo Simulations

Different Scenes

Using simulators, changing scenes is much easier

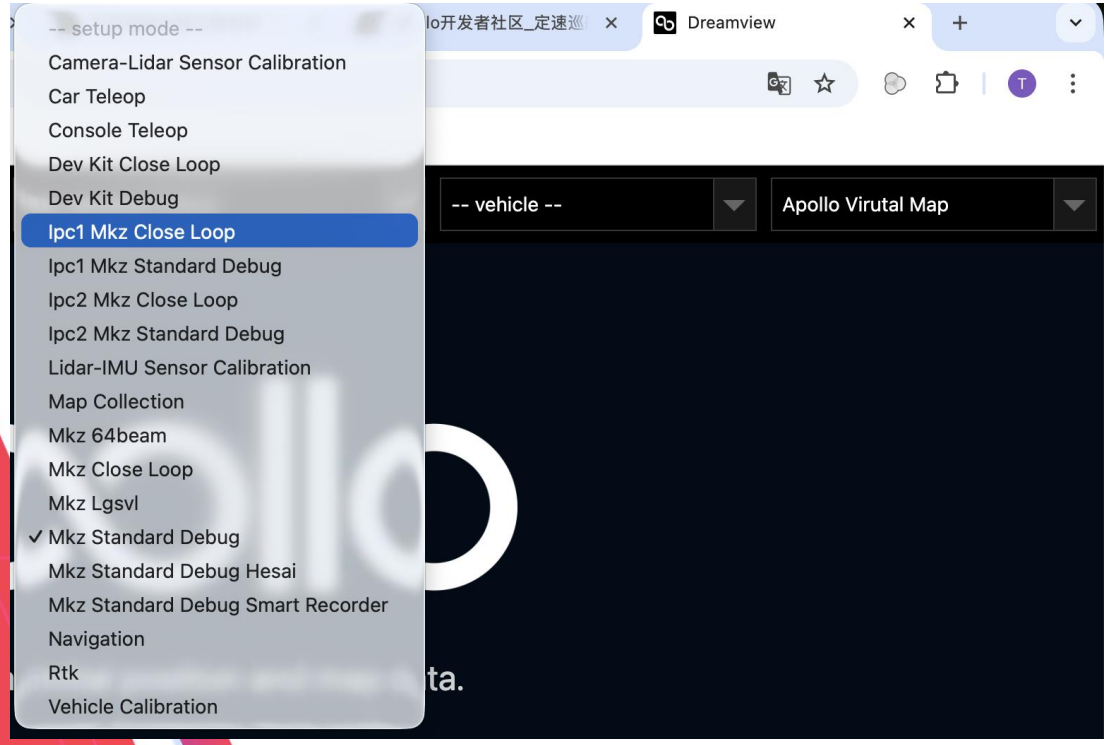
Sensors and Groundtruth

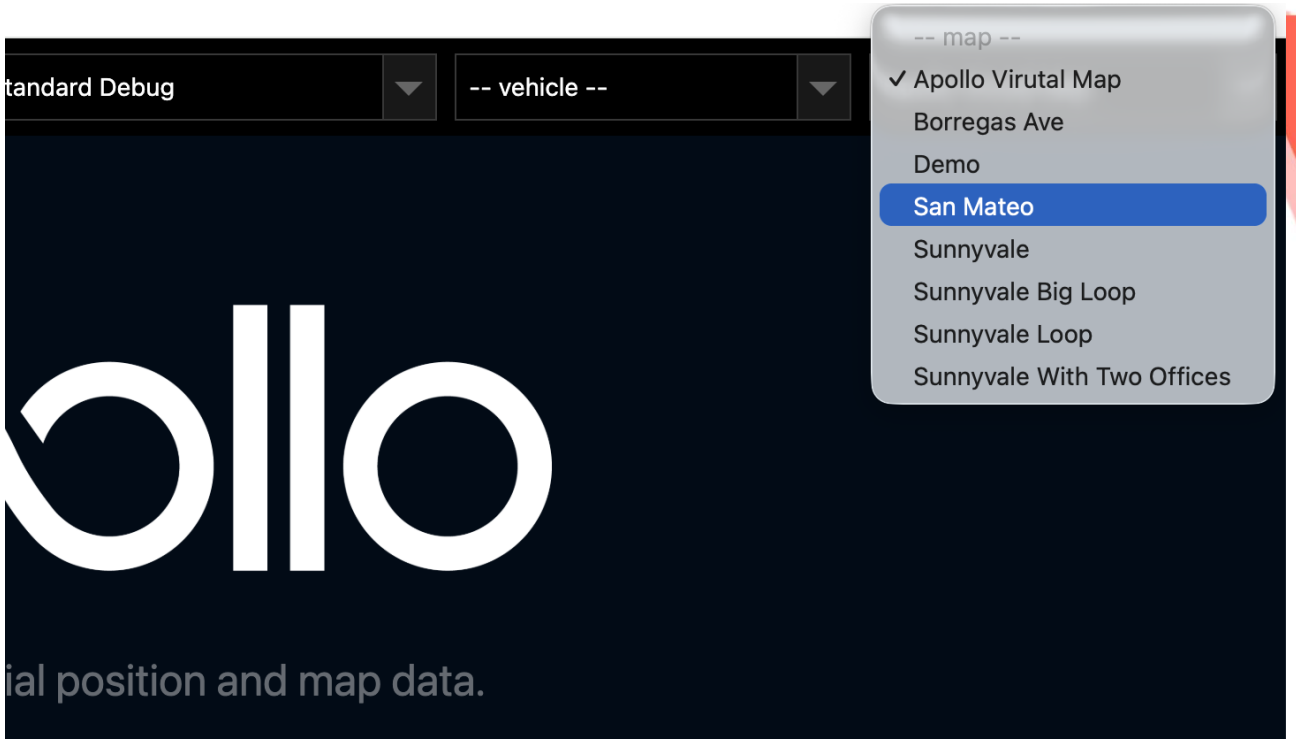
Multiple sensors provided and automated label generation

Apollo Simulator

Apollo Simulator

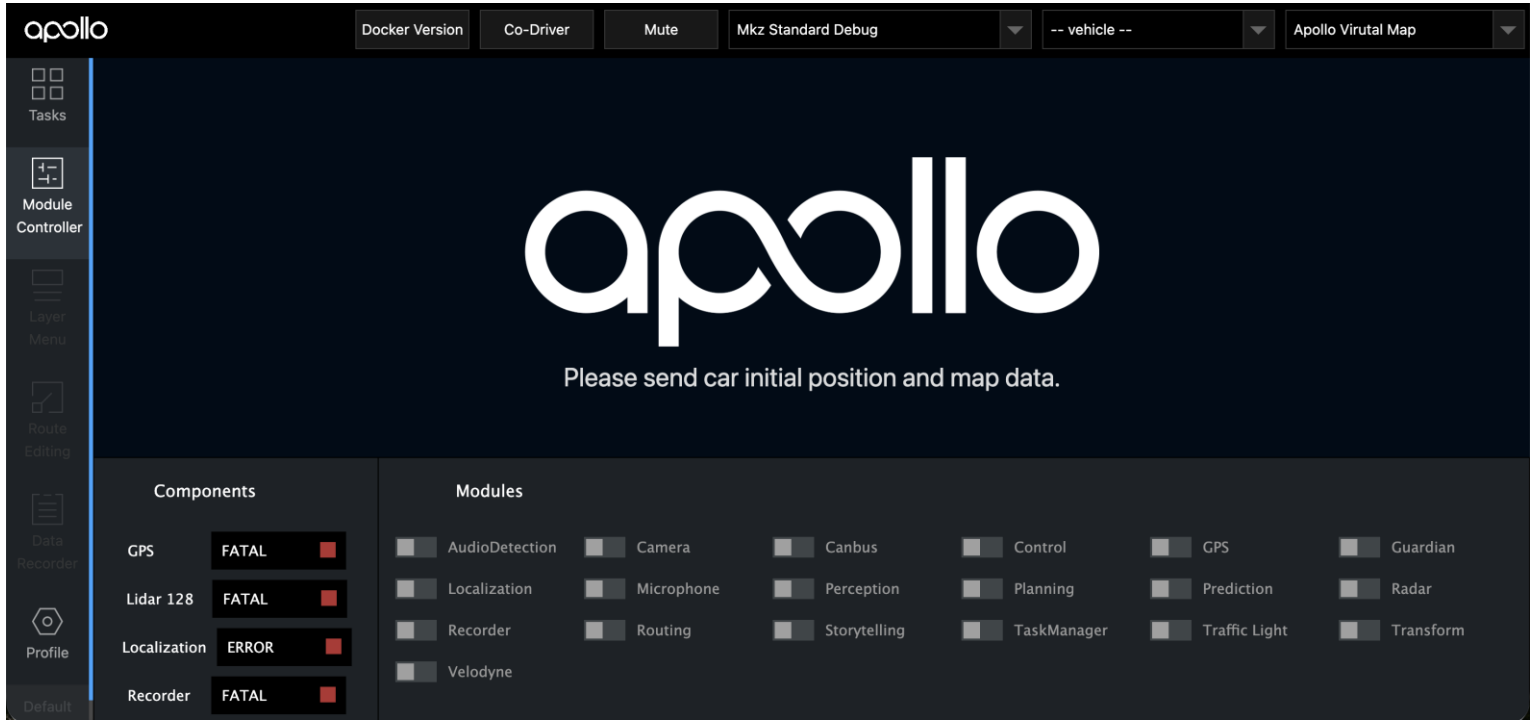
- Scenarios
  - Allows user to input different road types, obstacles, driving plans and traffic light states

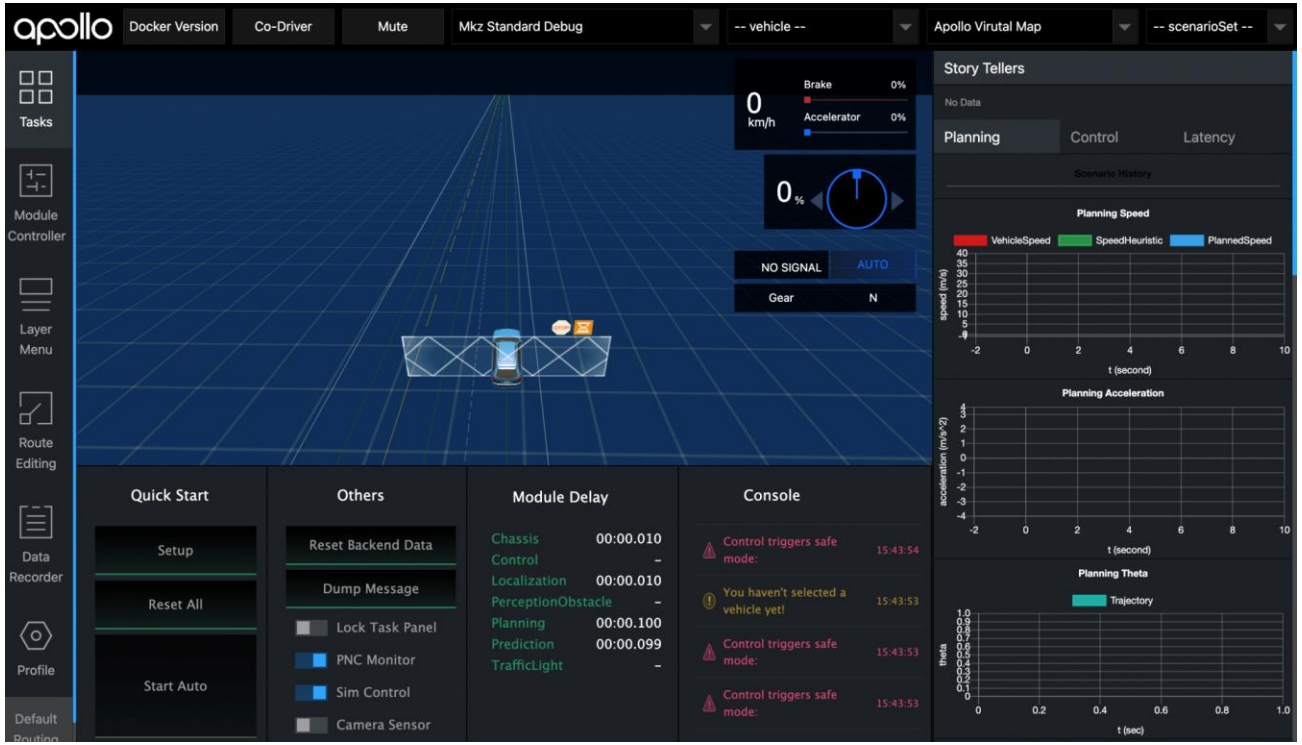




Apollo Simulations

- Execution modes
  - Gives users a complete setup to run multiple scenarios, and upload and verify modules in the Apollo environment



Apollo Simulations

- 3D visualization
  - Illustrates real-time road conditions and visualizes module output, while showing the status of the autonomous vehicle